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Final Report

The Art of Scientific Computation: Power Laws and Finite Size Scaling in the Abelian Sandpiles Model

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1

Introduction

The concept of self-organised criticality was first introduced by Bak, Tang, and Wiesenfeld in 1987 [1]. The central idea is that some slowly-varying complex systems drive themselves towards critical attractor states over time. In these states, consequences of a tiny perturbation are scale invariant. This means that colossal events can be triggered by miniscule changes to the system; such events may often follow a power law distribution. Many observed physical phenomena obey power law scaling - including earthquakes, forest fires, epidemics, etc. However, there is often a physical limit to the size of the event, which we characterise by the constant L.

The goal of this project is to simulate a complex system which evolves based only on rules connecting small units, and study the distributions of resulting system-level phenomena. To do so, we implement the Abelian (BTW) Sandpiles Model discussed in [1]. This model is formalised below, but in short, a grain of sand is randomly deposited on a square $L \times L$ lattice at each step of the simulation, and adds to the height of the sandpile at the site where it is deposited. When a site on the lattice reaches a critical height, it topples, distributing its grains to its neighbouring sites. This can trigger a chain of topplings, called an avalanche.

The system can be analysed through a range of statistics. In this report, we focus on the avalanche size (number of topplings), the avalanche area (number of sites toppled), and the L_1 distance between the deposition site and the furthest site toppled in an avalanche.

The research question investigated in this report is: *How can we model the distribution of key avalanche statistics, including the power law coefficient and lattice size-dependent tail decay?* To do so, we model the distributions and finite lattice size effects explicitly, using the method of (marginal) maximum likelihood estimation to fit distribution parameters from the simulated model. Finally, we examine evidence supporting and opposing the finite-size scaling hypothesis, which supposes the form of L-dependence in the probability mass function, in each of the studied statistics.

1.1 Mathematical Formulation of the BTW Model

Here, we formally introduce the Abelian Sandpiles Model and argue why the methods used in this report are appropriate. Below, we adapt the notation and arguments of Járai, 2018 [5].

Begin by defining a two-dimensional square lattice, which can be thought of as a grid with integer rows and columns. With lattice size L, this is the graph $G_L = (V_L, E_L)$, where

$$V_L := \{(x, y) \in \mathbb{Z}^2 : 1 \leq x, y \leq L\}$$

defines the set of vertices and

$$E_L := \{\mathbf{xy} : \mathbf{x}, \mathbf{y} \in V_L \wedge \|\mathbf{x} - \mathbf{y}\|_1 = 1\}$$

defines the set of edges, where $\|\mathbf{x} - \mathbf{y}\|_1$ denotes the L_1 norm, i.e. $\|(x_1, x_2) - (y_1, y_2)\|_1 = |x_1 - y_1| + |x_2 - y_2|$.

Each cell in the grid connects with its vertical and horizontal neighbours. Let $N(\mathbf{x}) = \{\mathbf{y} \in V_L : \mathbf{xy} \in E_L\}$ be this neighbourhood of $\mathbf{x}$ in G.

A sandpile is a pile of grains sitting on the lattice, defined by a height at each cell. Formally, it is a map $\eta : V_L \to \mathbb{Z}_{\geq 0}$, where $\eta(\mathbf{x})$ represents the height of the sandpile at site $\mathbf{x}$.

In the BTW model, sandpiles cannot support sites that are too high. Define the critical height as $\eta_c := \max_{\mathbf{x} \in V_L} \deg_G(\mathbf{x})$. In the above setup, $\eta_c = 4$ provided $L \geq 3$. A site $\mathbf{x} \in V_L$ is called stable iff $\eta(\mathbf{x}) < \eta_c$ and is otherwise unstable. The sandpile η is stable iff it is stable at each site $\mathbf{x} \in V_L$.

Let $U_\eta = \{\mathbf{x} \in V_L : \eta(\mathbf{x}) \geq \eta_c\}$ be the set of unstable sites.

Let $\Psi_{G_L} = \mathbb{Z}_{\geq 0}^{V_L}$ be the set of possible sandpiles and let $\Omega_{G_L} = \{0, 1, 2, ..., \eta_c - 1\}^{V_L} \subseteq \Psi_{G_L}$ be the set of stable sandpiles.

In the BTW model, when site heights exceed the critical height, they topple and redistribute their grains to their neighbours. We represent toppling as a function $\phi : \Psi_{G_L} \times V_L \to \Psi_{G_L}$ mapping $(\eta, \mathbf{x}) \mapsto \eta'$ where

$$\eta'(\mathbf{y}) = \begin{cases} \eta(\mathbf{y}) + 1 & \mathbf{y} \in N(\mathbf{x}) \wedge \mathbf{x} \in U_\eta \\ \eta(\mathbf{y}) - \eta_c & \mathbf{y} = \mathbf{x} \in U_\eta \\ \eta(\mathbf{y}) & \text{otherwise} \end{cases}$$

Note that we allow open boundary conditions in this study. When a toppling is initiated at a boundary site, we lose grains from the system. The number of grains lost by toppling η at $\mathbf{x}$ is

$$\ell(\eta, \mathbf{x}) := \sum_{\mathbf{x} \in V_L} \eta(\mathbf{x}) - \sum_{\mathbf{x} \in V_L} \eta'(\mathbf{x}) \geq 0$$

.

Topplings can create additional unstable sites because of the grain redistribution effect, creating chains of topplings called avalanches, which propagate until all sites are stable. The avalanche (stabilising) function $\xi : \Psi_{G_L} \to \Omega_{G_L}$ maps $\eta = \eta^{(0)} \mapsto \eta^{(s)}$ where $\eta^{(s)}$ is the stable sandpile output by the following algorithm:

Algorithm: Avalanche Function
$i \leftarrow 0$
while $U_{\eta^{(i)}} \neq \emptyset$ **do**:
choose $\mathbf{x}^{(i+1)} \in U_{\eta^{(i)}}$
$\eta^{(i+1)} \leftarrow \phi(\eta^{(i)}, \mathbf{x}^{(i+1)})$
$i \leftarrow i + 1$
end while
return $\eta^{(i)} = \eta^{(s)}$

When $\eta^{(s)} \neq \eta$, we say that an avalanche has occurred. Figure 1.1 illustrates the Avalanche Function on an example sandpile.

Avalanches can be described by a range of properties:

1. The avalanche (toppling) size is the final index $s(\eta) = s$
2. The avalanche (toppling) area $a(\eta)$ is the size of the set $\{\mathbf{x}^{(i)} : i \in \{1, \cdots, s\}\}$
3. The loss is $\ell(\eta) := \sum_{i=1}^{s} \ell(\eta^{(i-1)}, \mathbf{x}^{(i)})$
4. The L_1 length is $d(\eta) := \max_{i \in \{1, \cdots, s\}} \left\|\mathbf{x}^{(1)} - \mathbf{x}^{(i)}\right\|_1$.

When an avalanche occurs, note the following constraints:

1. $0 < a(\eta) \leq s(\eta)$
2. $a(\eta) \leq L^2$
3. $0 \leq \ell(\eta) \leq \sum_{\mathbf{x} \in V_L} \eta(\mathbf{x})$
4. $0 \leq d(\eta) \leq 2L$

When no avalanche occurs, we define $s(\eta) = a(\eta) = \ell(\eta) = d(\eta) = 0$ for convenience. However, our focus in this project is on the distribution of these statistics where they are non-zero. For avalanche size and area, this is conditional on an avalanche occurring. For loss, this is conditional on at least one boundary site topple. For L_1 length, this is conditional on $a(\eta) \geq 2$.

A proof that ξ is well-defined and independent of the choice of $\mathbf{x}^{(j)}$ in the loop can be found in Járai, 2018 [5]. The abelian nature of the topplings were first shown by Dhar in 1990 [4].

1.2 Probabilistic Framework

Randomness in the model comes from how new grains are deposited on the lattice.

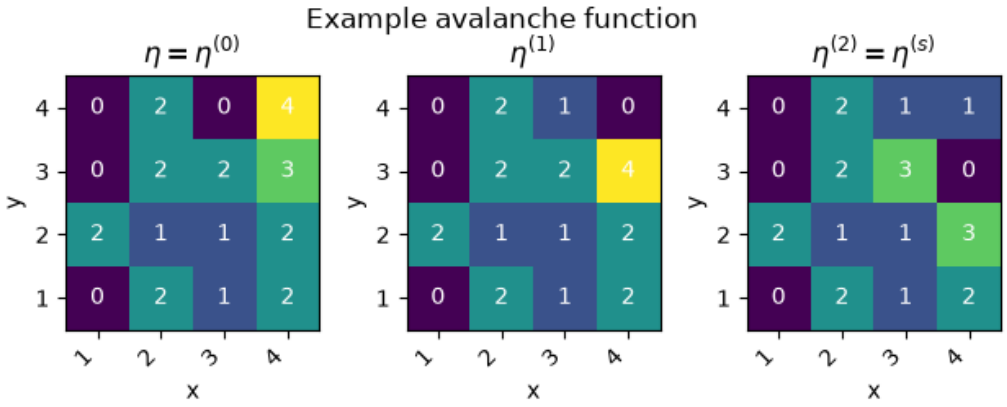


Figure 1.1: Example computation of $\xi(\eta)$ for an unstable sandpile with lattice size $L = 4$ and $\eta_c = 4$. Numbers written at each grid cell represent the sandpile height $\eta(x, y)$, and correspond to the cell colour. η is first toppled at site $\mathbf{x}^{(1)} = (4, 4)$ to produce $\eta^{(1)}$, which is also unstable. $\eta^{(1)}$ is subsequently toppled at site $\mathbf{x}^{(2)} = (4, 3)$ to produce the stable sandpile $\eta^{(2)} \in \Omega_{G_L}$. The algorithm terminates with $s(\eta) = 2$. In the above example, we have $a(\eta) = 2$, $\ell(\eta) = 2 + 1 = 3$, and $d(\eta) = 1$.

Define the deposition function $\delta : \Psi_{G_L} \times V_L \to \Psi_{G_L}$ as the map $\eta \mapsto \eta'$ with rule

$$\eta'(\mathbf{y}) = \begin{cases} \eta(\mathbf{y}) + 1 & \mathbf{y} = \mathbf{x} \\ \eta(\mathbf{y}) & \mathbf{y} \neq \mathbf{x} \end{cases}$$

Figure 1.2 illustrates the deposition function applied to an example sandpile and site.

We describe the sandpile evolution process by considering the sequence of iid random variables $\mathbf{X}_1, \mathbf{X}_2 ...$, each with support V_L and pmf

$$p(\mathbf{X}_j = \mathbf{x}) = L^{-2}$$

This represents a sequence of uniformly chosen random vertices, which are taken as locations for grain deposition. The model evolves as follows: a random grain is deposited on the existing sandpile, topplings are carried out until the resulting sandpile is stable, and the process repeats.

Mathematically, we build the sequence of stable sandpiles from the sequence $\{\mathbf{X}_j\}_{j\geq 1}$by taking $\eta_0 = \mathbf{0}$ and $\eta_j = \xi(\delta(\eta_{j-1}, \mathbf{X}_j))$ for $j \geq 1$.

Likewise, we obtain the sequence of avalanche sizes $\{S_j\}_{j\geq 1}$ where $S_j := s(\delta(\eta_{j-1}, \mathbf{X}_j))$. A similar construction yields sequences of avalanche areas, losses, and L_1 lengths.

The sequence $\{\eta_j\}_{j\geq 0}$ forms a discrete Markov Chain with finite state space Ω_{G_L}; the distribution of η_{j+1} depends only on the previous state η_j and the transition probability is determined by the distribution of $\mathbf{X}_j$.

Jarai [5] discusses the Markov structure in detail, and proves that the distribution of states η_j converges to a stationary distribution $\pi(\eta)$ as $j \to \infty$, where $\pi(\eta) > 0$ iff η is a recurrent state in the Markov Chain.

Consequently, the sequence of avalanche sizes $\{S_j\}_{j\geq 1}$ and the other avalanche statistic sequences, which are functions of the states η_j and the random vertices $\mathbf{X}_j$, also converge in distribution to stationary distributions as $j \to \infty$.

1.3 Statistical Models

This project builds and compares statistical models for the distribution of positive avalanche size, area, and L_1 length as truncated power laws. Three distribution families are fit and compared, which are discrete power law distributions inspired by distributions described in Clauset et al. (2009) [3]. All three have support $X \in \{1, \cdots, x_{max}\} \subseteq \mathbb{Z}_{>0}$.

Bounded Power Law	$p(x) \propto x^{-\tau}$
Exponentially Truncated Power Law	$p(x) \propto x^{-\tau} e^{-x/c}$
Stretched Exponentially Truncated Power Law	$p(x) \propto x^{-\tau} e^{-(x/c)^{\beta}}$

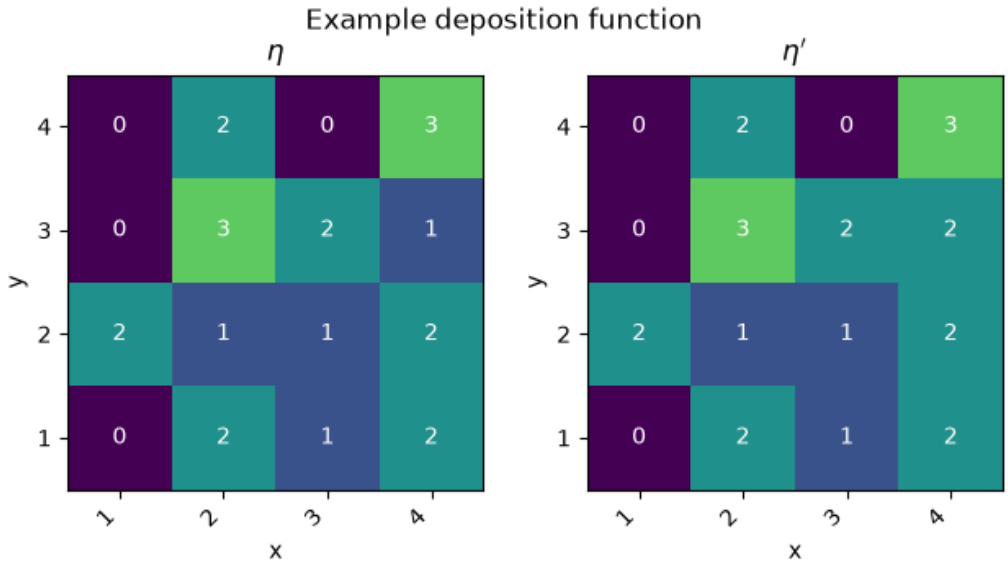


Figure 1.2: Example deposition function δ applied to a sandpile with lattice size $L = 4$. A grain is deposited at site $\mathbf{x} = (4, 3)$, increasing the site height from 1 to 2 and producing η'. Numbers written at each grid cell represent the sandpile height at vertex (x, y), and correspond to the cell colour.

Of particular interest is the finite scaling hypothesis described by Tebaldi et al. (1999) in [7], which hypothesises that the finite size lattice L influences the distribution via:

$$p(x) \propto x^{-\tau} f\left(\frac{x}{L^D}\right)$$

Here, D is the fractal dimension of the statistic. The truncated distributions above suggest forms for f, with $c = kL^D$.

As discussed in [3], estimating the power law coefficient using linear regression on the log-log plot of the pmf is not statistically justified, nor can it capture the finite size effect on the distribution tail. In this report, we apply the method of Maximum Likelihood Estimation recommended in [3], using exactly specified discrete distributions.

Particular care must be taken when performing inference because the samples collected are not independent. While Markov Chain stationarity supports the distribution of the observed samples approximating their stationary distribution, this does not mean that samples can be treated as independent and identically distributed samples from the stationary distribution. Consequently, standard goodness of fit tests, tests of statistical significance, and most quantitative comparison methods become computationally complex and highly approximate. Fortunately, maximum likelihood estimation is still justifiable. While it is not technically maximum likelihood, since dependence means that the likelihood cannot be written as the product of marginal pmfs, we will still refer to it as such in this report.

2

Simulation

2.1 Method

The stationarity of the avalanche statistic sequences forms the theoretical foundation for empirical simulation and analysis according to the following algorithm:

Algorithm: Simulating the Sandpile Model
initialise $\mathbf{s}, \mathbf{a}, \ell, \mathbf{d}$ as empty arrays of size n
initialise $\eta = \mathbf{0}$
for $i \in \{1, \cdots, m\}$ **do**:
simulate $\mathbf{X}_i$
$\eta \leftarrow \delta(\eta, \mathbf{X}_i)$
$\eta \leftarrow \xi(\eta)$
end for
for $i \in \{m+1, \cdots, m+n\}$ **do**:
simulate $\mathbf{X}_i$
$\eta \leftarrow \delta(\eta, \mathbf{X}_i)$
$\mathbf{s}[i-m] \leftarrow s(\eta)$
$\mathbf{a}[i-m] \leftarrow a(\eta)$
$\ell[i-m] \leftarrow \ell(\eta)$
$\mathbf{d}[i-m] \leftarrow d(\eta)$
$\eta \leftarrow \xi(\eta)$
end for
return $\mathbf{s}, \mathbf{a}, \ell, \mathbf{d}$

We can then discard the zero values of the returned arrays, because our interest is in modelling the distributions of the statistics given they are nonzero. In the case of avalanche size and area, this is conditional on an avalanche occurring.

In the above, simulated states $\eta_1, \cdots, \eta_m$ are called burn-in states, and are discarded. After this period, we assume that the distribution of η_j is approximately stationary. Therefore, simulations $\eta_{m+1}, \cdots, \eta_{m+n}$ are retained and their empirical distribution approximates the stationary distribution $\pi(\eta)$. From that sequence, we construct simulated sequences for each avalanche statistic of length n, whose distribution approximates its respective stationary distribution.

2.2 Practicalities

In practice, hundreds of thousands of simulations are required to approach stationarity. This is computationally expensive, especially for large lattice sizes, because computations cannot be parallelised due to the Markov structure of the problem. An initial Python implementation of the simulation procedure above was too slow to produce meaningful results. The final implementation uses `numba` with `njit` decorators, which stands for 'no python just in time' [6]. This compiles the Python code prior to execution, bypassing the CPython interpreter and achieving execution speed near code written in C. However, only a subset of Python and `numpy` functionality is permitted, requiring significant refactoring away from object-oriented design. Table 2.1 compares identical

L	`numba`-optimised runtime (s)	standard runtime (s)	speedup
5	0.03	3.90	135.15
7	0.05	5.93	117.16
10	0.09	9.41	104.69
12	0.12	11.68	99.98
15	0.13	17.62	132.55

Table 2.1: Runtime comparison between the `numba`-based implementation and the standard Python implementation. The optimised code runs 100-140x faster. Simulations comprised 100,000 depositions (40,000 burn-in and 60,000 subsequent) and were performed on lattice sizes $L \in \{5, 7, 10, 12, 15\}$.

implementations of the simulation for varying lattice sizes, where one uses `numba @njit` decorators, while the other does not. Results show a 100-140x speedup using the optimised code.

The original implementation used a Python in-built queue structure to keep track of pending topples. However, this was not supported by `numba`. Instead, the implementation uses a circular queue data structure using a numpy array of coordinates. One index points to the head of the queue, and another points to the tail. This allows $O(1)$ enqueue and dequeue operations. Sites are only added to the queue if their height is exactly 4, as this prevents needing to search through the queue to avoid duplicates. The assumption required is that the sandpile is stable before deposition, which is correct in this simulation. All sandpile statistics are recorded in $O(1)$ time for each topple in the avalanche. In particular, avalanche area uses a duplicate board to record the last avalanche in which each site was toppled. This allows for $O(1)$ lookup to see if a site is newly toppled in a given avalanche. Therefore, the avalanche function implemented runs in $O(s)$ time and requires $O(L^2)$ space.

2.3 Results

Simulations were performed for lattice sizes $L \in \{5, 10, 15, 20, 30, 40, 50, 65, 80, 100\}$. The burn-in period was set to $m = 4 \times 10^4$ and sequences of length $n = 1 \times 10^6$ were retained for each lattice size. This burn-in period was determined somewhat arbitrarily - it corresponds to $\eta_c L^2$ for the largest simulated lattice size. This is a rule-of-thumb approach that we assume is sufficient.

The plots in Figure 2.1 depict a linear trend in early values of avalanche size, area, and L_1 distance, suggesting a power law distribution followed by a sharp decay. The distribution for Loss does not appear to have this power law structure, and the remainder of this report focusses on understanding and analysing the distribution of avalanche size, area, and length.

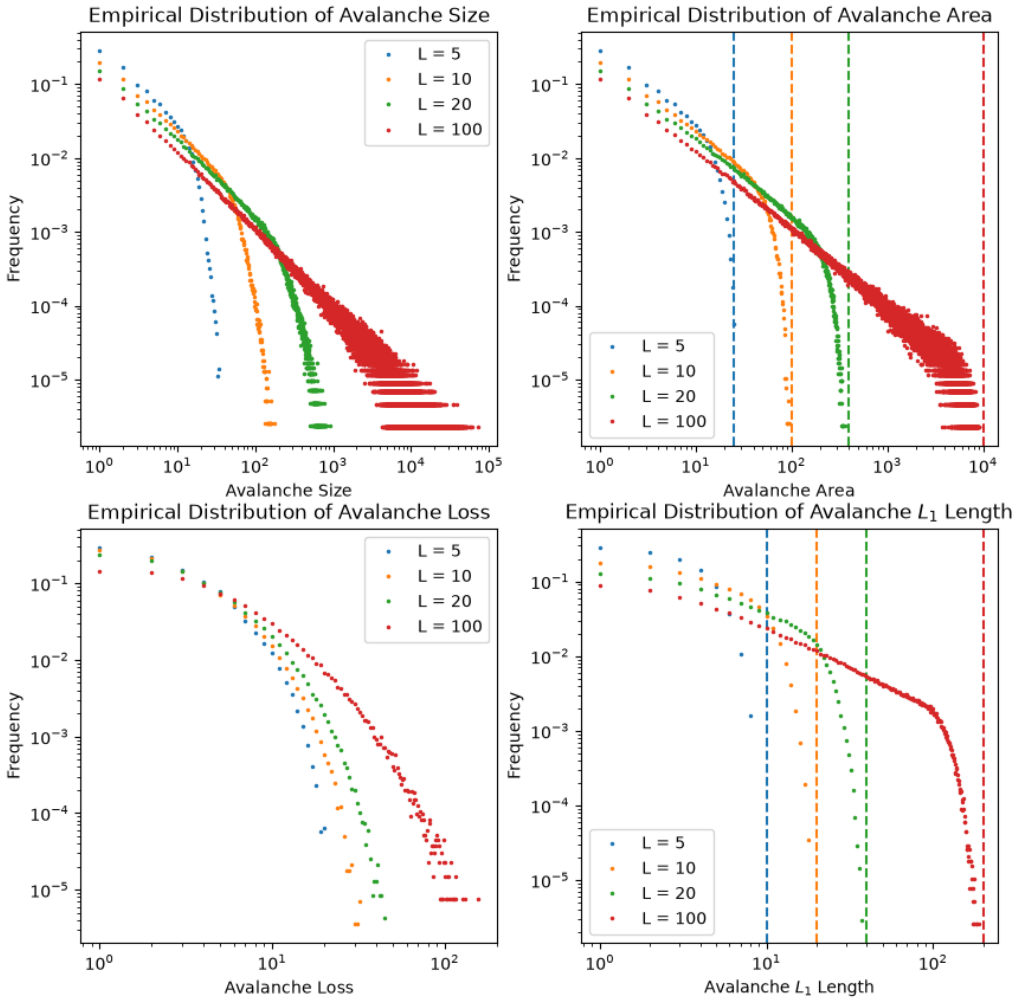


Figure 2.1: Observed frequencies of key avalanche statistics computed from the simulated sandpile Markov chain, excluding the discarded burn-in samples. A subset of four lattice sizes are depicted: 5, 10, 20, and 100, with varying colors. Values are computed from 10^6 simulated observations. The dashed vertical lines on the Avalanche Area and L_1 Length plots correspond to L^2 and $2L$ respectively. Plots are shown in log-log axes. The appearance of horizontal lines is an artifact of the log-log axes. The lines correspond to sizes with only 1, 2, 3, etc. observations in the dataset, with exaggerated vertical distance between the lines because of the log scale.

3

Power Law Fit

3.1 Bounded Power Law Fit

The linear component of the distribution observed in Figure 2.1 suggests fitting a power law distribution to the simulated sequence of avalanche sizes and areas.

We hypothesize a power law scaling relationship for Toppling Size S and area A of the form:

$$p(X = x; \tau, x_{max}) = \frac{1}{C_{\tau,x_{max}}} x^{-\tau} \text{ for } x \in \{1, \cdots, x_{max}\}$$

where $x_{max} = x_{max}(L)$ is a cutoff value reflecting the observed decay of the distribution away from power law scaling due to the finite lattice size L.

The shape of the pmf under varying parameter values can be observed in Figure 3.1.

To estimate the power law scaling coefficient τ, we apply the method of Maximum Likelihood as justified in Section 1.3.

Let $\mathbf{x} = (x_1, x_2, \cdots, x_n)$. The likelihood function is:

$$\mathcal{L}(\mathbf{x}; \tau, x_{max}) = \prod_{i=1}^{n} \frac{x_i^{-\tau}}{C_{\tau,x_{max}}}$$

$$\log \mathcal{L}(\mathbf{x}; \tau, x_{max}) = -n \log C_{\tau,x_{max}} - \tau \sum_{i=1}^{n} \log x_i$$

Where $C_{\tau,x_{max}}$ is the normalising constant:

$$C_{\tau,x_{max}} = \sum_{k=1}^{x_{max}} k^{-\tau}$$

To estimate the parameter τ assuming known x_{max}, we compute its Maximum Likelihood Estimate (MLE). This is the value $\hat{\tau}$ that satisfies:

$$\hat{\tau} = \arg\max_{\tau} \mathcal{L}(\mathbf{x}; \tau, x_{max}) = \arg\min_{\tau} \{-\log \mathcal{L}(\mathbf{x}; \tau, x_{max})\}$$

A practical note: Because the normalising constant $C_{\tau,x_{max}}$ depends on the parameter τ, we cannot find an analytic solution for $\hat{\tau}$ and rely on numerical optimisation techniques. The numerical optimisation technique used here is the L-BFGS-B algorithm introduced by Byrd et al. (1995) [2] and implemented in the `scipy` python library. Vectorisation is used via `numpy` arrays to improve the efficiency of the likelihood estimation, which requires computing the log-pmf of each sample point. Finally, to prevent numerical underflow when dealing with tiny probabilties of tail events, the `logsumexp` function from `scipy` was used; this function provides an underflow-resistant method for calculating an expression of the form $\log \sum_i e^{Q_i}$ where e^{Q_i} ranges across a range of scales, which was the case when computing the normalising constants. These methods were used across all likelihood estimations in this project.

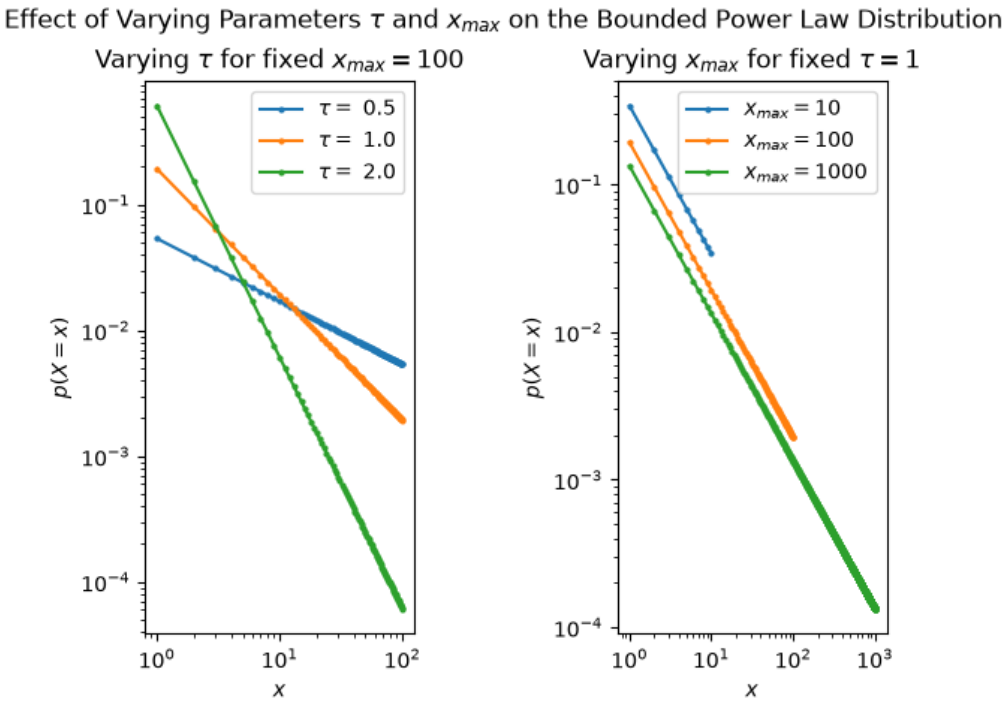


Figure 3.1: Visualisation of the bounded power law probability mass function on log-log axes under different parameter combinations. Each point represents a discrete value and corresponding probability mass, with a line connecting sequential points for illustration purposes. The left plot shows the effect of varying τ while keeping x_{max} constant, which results in changes to the slope and intercept of the connecting line. The right plot shows the effect of varying x_{max} while keeping τ constant, which results in vertical translation of the connecting line, with different support for x. The orange line is identical in both plots, corresponding to parameters $\tau = 1$, $x_{max} = 100$.

Let $\tau = \tau_s(L), x_{max} = s_{max}(L)$ for modelling the avalanche size, $\tau = \tau_a(L), x_{max} = a_{max}(L)$ for modelling the avalanche area, and $\tau = \tau_d(L), x_{max} = d_{max}(L)$ for modelling the avalanche L_1 length. We first consider a full-data estimation procedure, where we ignore the non-linear distribution tails and fit τ for each statistic considered and lattice size simulated. Here, we fix $a_{max}(L) = L^2$, $d_{max}(L) = 2L$, and $s_{max}(L) = 10L^2$ as known; this is clearly justified as the theoretical maximum for avalanche area and a conservative maximum for L_1 length, and is a computational adjustment for avalanche size that exceeds any observed avalanche size in the simulated data.

Under the finite size scaling hypothesis discussed in 1.3, we expect that the power law coefficient should be constant for all values of L, i.e. we expect $\tau_s(L) = \tau_s$, $\tau_a(L) = \tau_a$, and $\tau_d(L) = \tau_d$.

The estimates obtained are presented in Figure 3.2 and Table 3.1. We observe a decreasing trend in the estimated power law coefficients for both avalanche size and avalanche area. However, in Figure 3.3 we see that the tail has a substantial effect on the fit, creating an artificially high estimate for the slope. This effect appears most pronounced at smaller lattice sizes, but is still evident at $L = 100$, suggesting the decrease in $\hat{\tau}$ could be caused by poor fit at the tails.

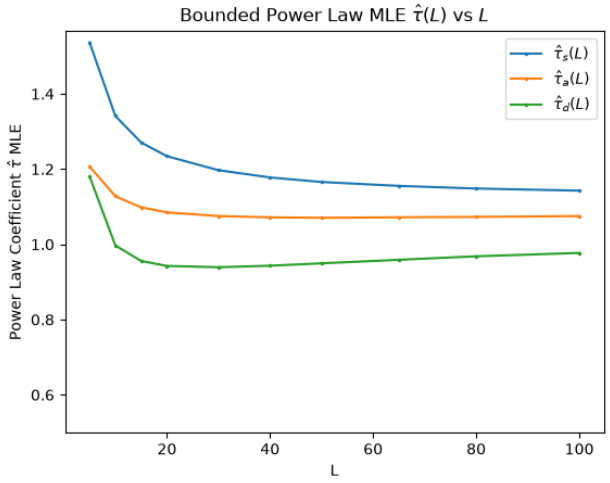


Figure 3.2: Estimates for $\hat{\tau}_s(L), \hat{\tau}_a(L), \hat{\tau}_d(L)$.

L	$\hat{\tau}_s(L)$	$\hat{\tau}_a(L)$	$\hat{\tau}_d(L)$
5	1.54	1.21	1.18
10	1.34	1.13	1.00
15	1.27	1.10	0.96
20	1.23	1.09	0.94
30	1.20	1.08	0.94
40	1.18	1.07	0.94
50	1.17	1.07	0.95
65	1.16	1.07	0.96
80	1.15	1.07	0.97
100	1.14	1.08	0.98

Table 3.1: Bounded Power Law MLE coefficients $\hat{\tau}$ obtained for varying lattice sizes L

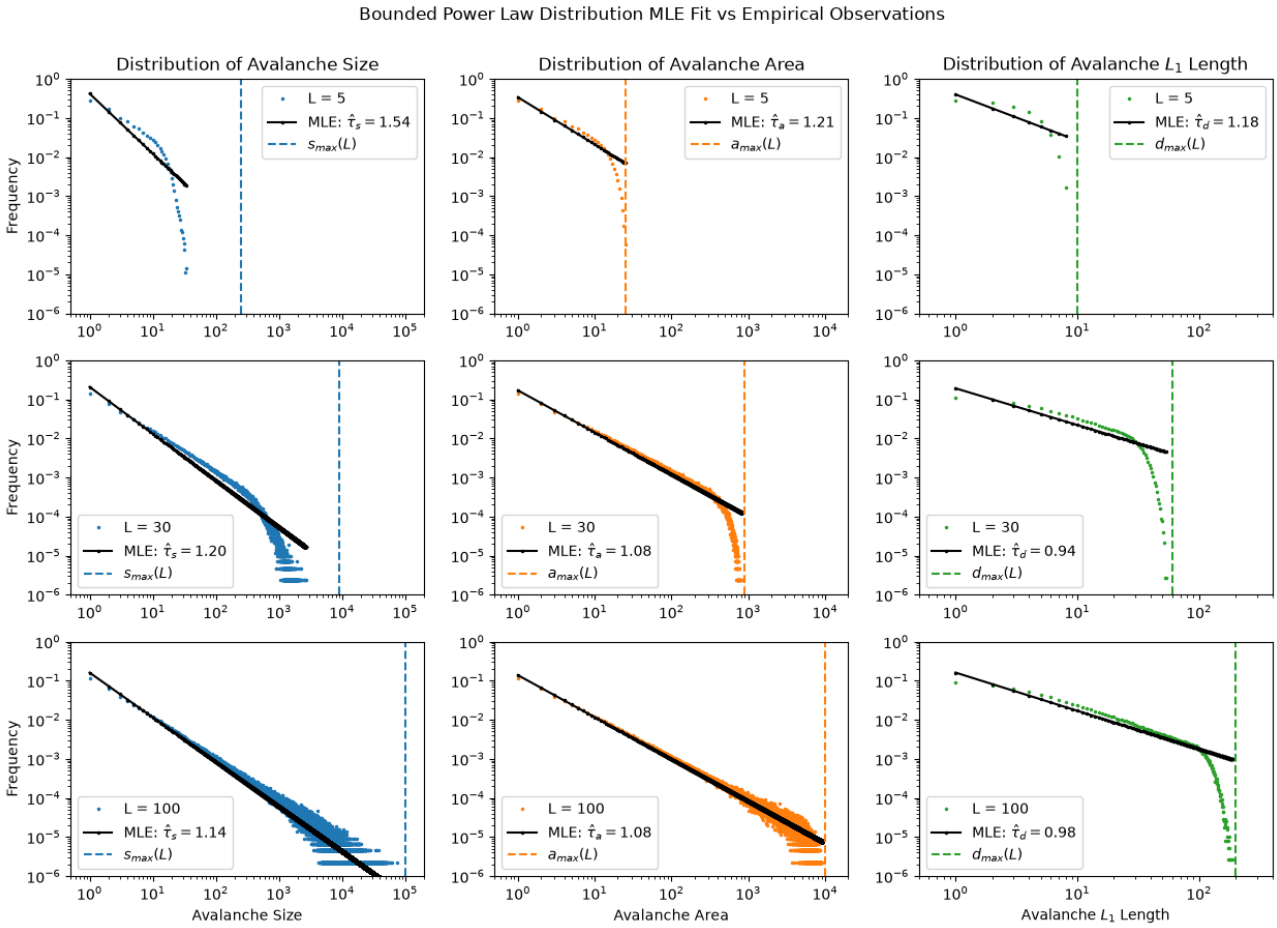


Figure 3.3: Full-data bounded power law MLE fits overestimates the slope relative to the linear section of the data. Select lattice sizes $L = 5, 30, 100$ are depicted. The fits for all lattice sizes are depicted in Appendix Figure A.1. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, while the right column shows avalanche L_1 length. The dashed vertical lines represent the fixed parameter x_{max}.

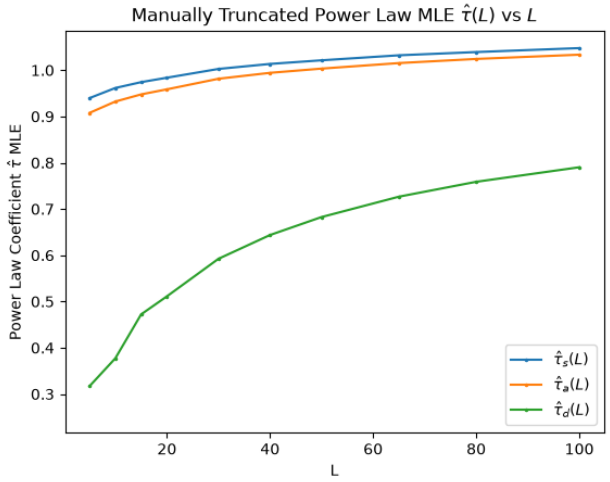


Figure 3.4: Estimates for $\hat{\tau}_s(L), \hat{\tau}_a(L), \hat{\tau}_d(L)$ under the manual truncation rule $s_{max} = 0.2L^2, a_{max} = 0.1L^2, d_{max} = 0.5L$.

L	$\hat{\tau}_s(L)$	$\hat{\tau}_a(L)$	$\hat{\tau}_d(L)$
5	0.94	0.91	0.32
10	0.96	0.93	0.38
15	0.97	0.95	0.47
20	0.98	0.96	0.51
30	1.00	0.98	0.59
40	1.01	0.99	0.64
50	1.02	1.00	0.68
65	1.03	1.02	0.73
80	1.04	1.02	0.76
100	1.05	1.03	0.79

Table 3.2: MLE power-law slopes obtained for varying lattice sizes L under the manual truncation rule $s_{max} = 0.2L^2, a_{max} = 0.1L^2, d_{max} = 0.5L$

3.2 Manually-truncated Power Law

As a first correction for the finite lattice effect, we consider restricting $s_{max}, a_{max}, d_{max}$ to include only those topple sizes, areas, and lengths for which linearity is observed in the log-log plot. As we saw in Figure 3.1, changing x_{max} should not affect the estimated slope of the distribution.

To avoid manually setting an arbitrary maximum for each value of L, here we consider the rule-of-thumb approximation $s_{max} = 0.2L^2, a_{max} = 0.1L^2, d_{max} = 0.5L$. Clearly, there are significant limitations to this approach: the cutoff rule was chosen arbitrarily, and data is significantly limited for lower values of L. However this will help us get a more accurate idea of the power law structure.

The results in Table 3.2 and Figure 3.4 are noticeably lower than the previous estimates in Table 3.1, with an increasing trend rather than decreasing. The fits in Figure 3.5 are reasonably good for avalanche size and area, however nonlinearity is visible in the avalanche L_1 distance plot. This may indicate that the assumed power law scaling is not as well supported from the obtained distribution as it is for size and area.

From these two analyses, it is reasonable that the powerlaw-governed section of the distribution has exponent $\tau_s, \tau_a \approx 1$, which is consistent with the estimate presented in the original 1987 paper by Bak et al. [1].

To avoid discarding data, and to model the distribution explicitly, we turn to models that allow for tail decay. We attempt to capture the effect of the finite lattice size L and obtain a more informed approximation to the power law scaling coefficient.

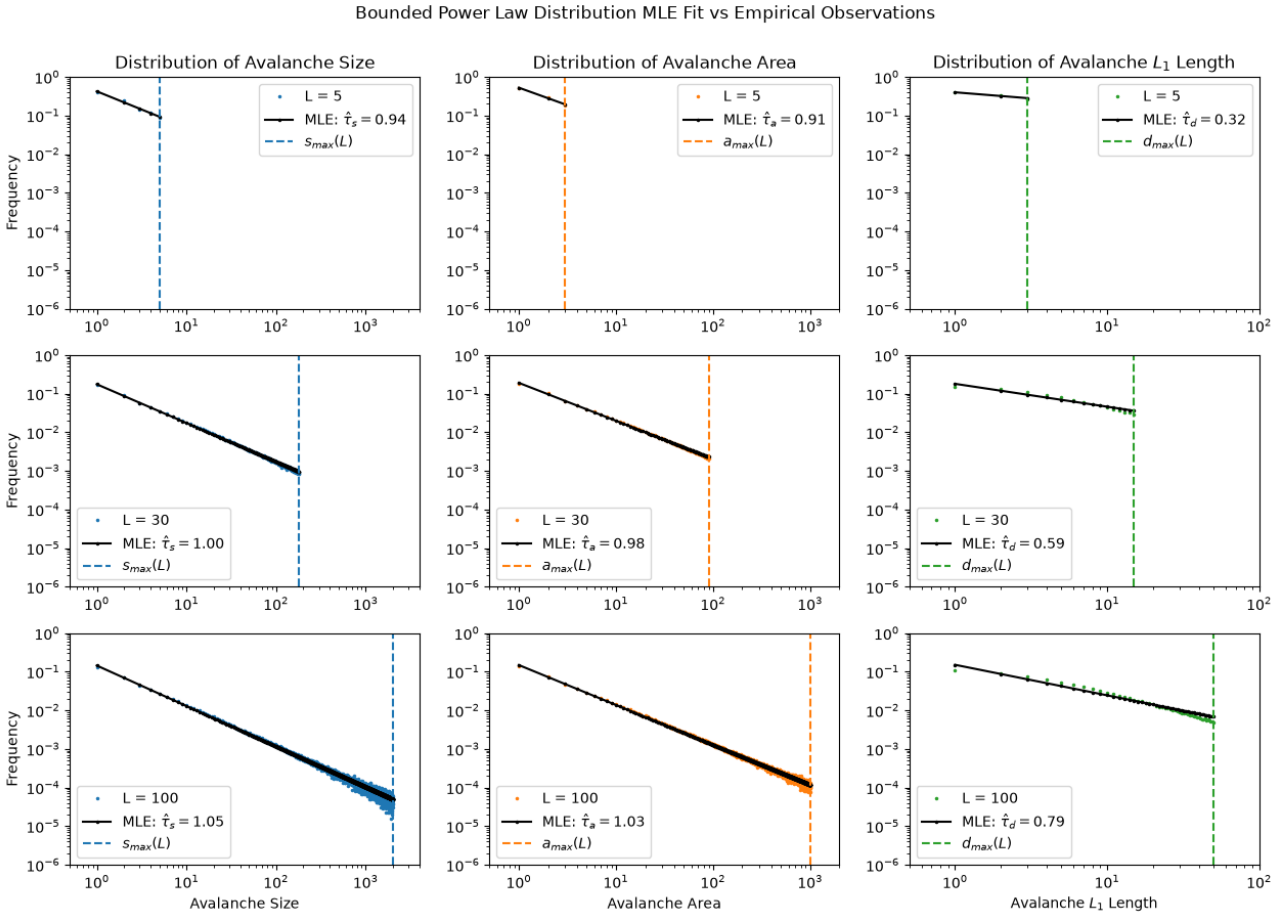


Figure 3.5: MLE Estimates of the power law scaling coefficient obtained by manually truncating the data tail. Lattice sizes $L = 5, 30, 100$ are depicted - the full data fits are shown in Appendix Figure A.2. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, and the right column shows avalanche L_1 length. The dashed vertical lines on the right show the manual truncation rule bounds $s_{max} = 0.2L^2, a_{max} = 0.1L^2, d_{max} = 0.5L$.

4

Finite Size Scaling Effect

Recalling Section 1.3, the finite size scaling law hypothesizes that the tail behaviour relates to the finite lattice size L via a probability mass function of the form:

$$p(X = x; \tau, L, f, D) \propto x^{-\tau} f\left(\frac{x}{L^D}\right)$$

Here f governs the tail behaviour of the distribution, i.e. for small values of u, $f(u) \approx 1$ and the distribution resembles a power law.

We first model the distribution separately for each lattice size L, obtain estimates, assess the fits, and compare these to pooled estimates obtained by relating the pmf form to the finite scaling function f above.

4.1 Exponentially Truncated Power Law

We first consider the exponentially-truncated power law:

$$p(X = x; \tau, c, x_{max}) = \frac{1}{C_{\tau,c,x_{max}}} x^{-\tau} \exp\left(\frac{-x}{c}\right)$$

Where $C_{\tau,c,x_{max}}$ is the normalising constant:

$$C_{\tau,c,x_{max}} = \sum_{k=1}^{x_{max}} k^{-\tau} \exp\left(-k/c\right)$$

In the above pmf, τ and x_{max} are, as before, the power law scaling coefficient and the support upper bound respectively, while the new parameter c controls the location of the soft cutoff, where power law behaviour decays. The shape of the pmf under varying parameter values can be observed in Figure 4.1

The likelihood and log-likelihood are:

$$\mathcal{L}(\mathbf{x}; \tau, c, x_{max}) = \prod_{i=1}^{n} \frac{1}{C_{\tau,c,x_{max}}} x_i^{-\tau} \exp\left(\frac{-x_i}{c}\right)$$

$$\log \mathcal{L}(\mathbf{x}; \tau, c, x_{max}) = -n \log C_{\tau,c,x_{max}} - \tau \sum_{i=1}^{n} \log x_i - \frac{1}{c} \sum_{i=1}^{n} x_i$$

Let $\mathbf{x}^j = (x_1^j, \cdots, x_n^j)$ be the set of simulated statistics from lattice size L_j. As before, for each L, we numerically optimise the log-likelihood function to obtain:

$$\hat{\tau}(L_j), \hat{c}(L_j) = \arg\min_{\tau,c} \left\{ -\log \mathcal{L}(\mathbf{x}^j; \tau, c, x_{max}) \right\}$$

Set $\tau = \tau_s(L), x_{max} = s_{max}(L), c = c_s(L)$ for modelling the avalanche size, $\tau = \tau_a(L), x_{max} = a_{max}(L), c = c_a(L)$ for the avalanche area, and $\tau = \tau_d(L), x_{max} = d_{max}(L), c = c_d(L)$ for modelling the avalanche length. Recall we fix $a_{max}(L) = L^2$, $d_{max} = 2L$, and $s_{max}(L) = 10L^2$ as known.

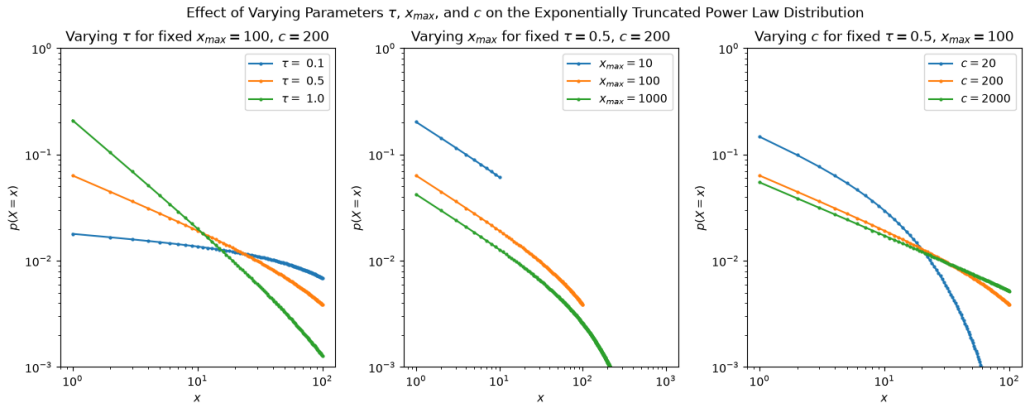


Figure 4.1: Visualisation of the exponentially truncated power law probability mass function on log-log axes under different parameter combinations. Each point represents a discrete value and corresponding probability mass, with a line connecting sequential points for illustration purposes. The left plot shows the effect of varying τ while keeping x_{max} and c constant, which results in changes to the slope of the linear part of the connecting line. The central plot shows the effect of varying x_{max} while keeping τ and c constant, which results in vertical translation of the connecting curve, with different support for x. The right plot shows the effect of increasing c while keeping τ and x_{max} constant, which results in later decay. The orange line is identical in all plots, corresponding to parameters $\tau = 0.5$, $x_{max} = 100$, and $c = 200$.

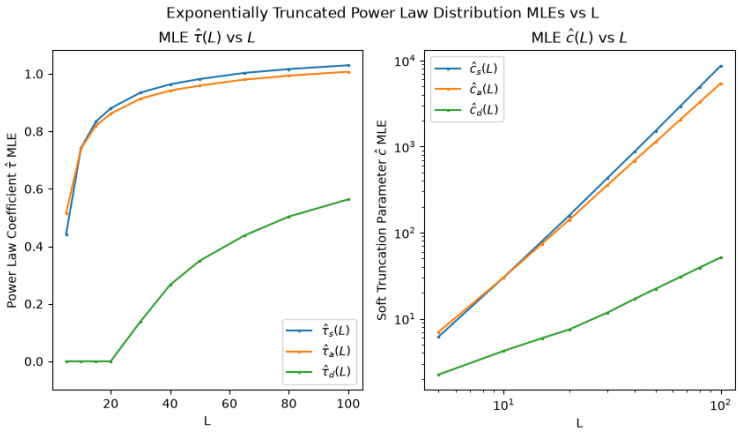


Figure 4.2: Estimates MLEs $\hat{\tau}_s(L), \hat{\tau}_a(L), \hat{\tau}_d(L)$ (left) and $\hat{c}_s(L), \hat{c}_a(L), \hat{c}_d(L)$ (right) for varying lattice sizes L.

L	$\hat{\tau}_s(L)$	$\hat{c}_s(L)$	$\hat{\tau}_a(L)$	$\hat{c}_a(L)$	$\hat{\tau}_d(L)$	$\hat{c}_d(L)$
5	0.44	6.15	0.52	7.01	0.00	2.24
10	0.74	30.26	0.74	30.24	0.00	4.22
15	0.83	79.33	0.82	74.17	0.00	5.93
20	0.88	157.56	0.86	140.37	0.00	7.47
30	0.93	428.18	0.91	355.29	0.14	11.73
40	0.96	868.82	0.94	683.27	0.27	16.88
50	0.98	1509.00	0.96	1124.60	0.35	22.16
65	1.00	2924.10	0.98	2041.31	0.44	30.43
80	1.02	4932.09	0.99	3269.35	0.50	39.24
100	1.03	8647.45	1.01	5427.84	0.56	51.29

Table 4.1: MLE power-law slopes τ and soft cutoffs c obtained for varying lattice sizes L

The results in Table 4.1 are promising: for lattice sizes 40 and above, the powerlaw coefficient $\tau \approx 1$. Further, in Figure 4.2 the log soft cutoffs $\log \hat{c}$ appear to scale linearly in $\log L$, a relationship that we will examine in more detail when computing the pooled estimators.

The MLE fits in Figure 4.3 appear reasonable for avalanche size in large L, however they do not appear to otherwise capture the full tail behaviour of the distributions of the other statistics or for smaller lattice sizes. Nevertheless, we proceed with the pooled fits.

4.2 Pooled Exponentially Truncated Power Law

Consider again the Finite Size Scaling hypothesis:

$$p(X = x; \tau, L, f, D) \propto x^{-\tau} f\left(\frac{x}{L^D}\right)$$

In the previous section, we modelled

$$p(X = x; \tau, L, c, D) \propto x^{-\tau} \exp\left(-\frac{x}{c}\right)$$

Therefore, if the chosen pmf is adequate to model the behaviour of the system, and the statistics obey finite

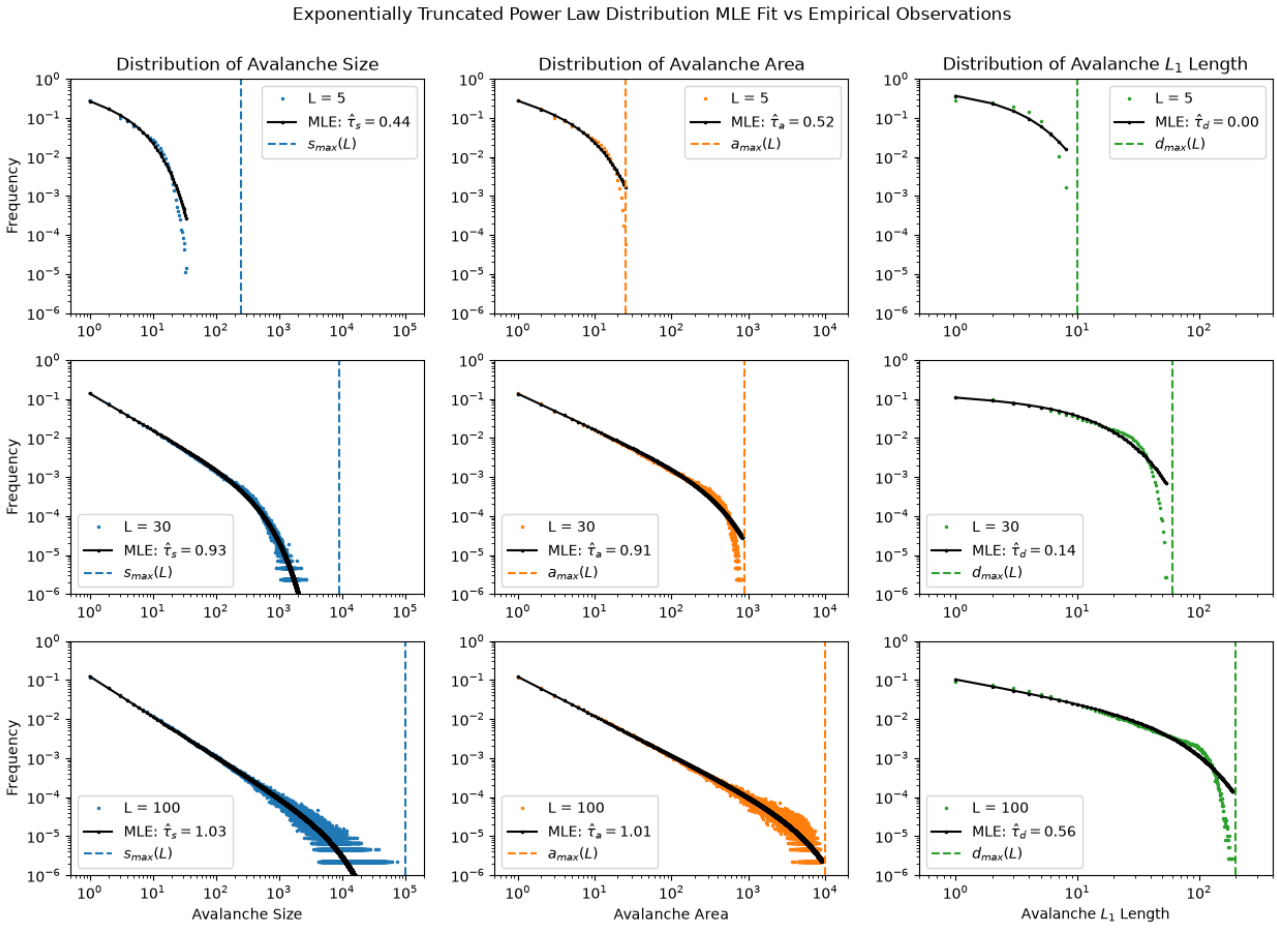


Figure 4.3: The MLE-fit exponential truncation parameter c partially captures the distribution tails, and performs well for avalanche size at higher L. Select lattice sizes $L = 5, 30, 100$ are depicted - the fits for all simulated lattice sizes are shown in Appendix Figure A.3. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, and the right column shows avalanche L_1 distance. The dashed vertical lines show the fixed parameter x_{max}.

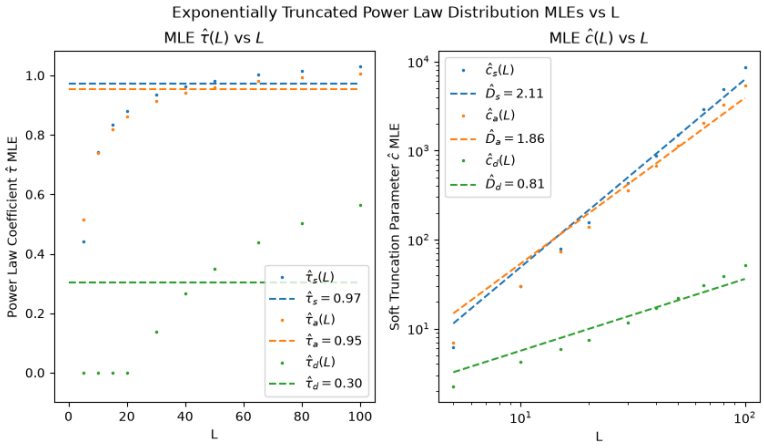


Figure 4.4: Unpooled MLE estimates of the parameters (points) vs pooled MLE parameter estimates (dashed lines). If the pmf form and the finite size scaling hypothesis are appropriate, we expect the dashed line to approximate the trend of the points. Based on the above, the model is not quite appropriate, and we need to consider an additional model for the pmf.

	$\hat{\tau}$	$\hat{k}$	$\hat{D}$
Size s	0.97	0.38	2.11
Area a	0.95	0.74	1.86
L_1 Length d	0.30	0.88	0.81

Table 4.2: Pooled MLE coefficients for finite size scaling assuming the exponentially truncated power law distribution. We expect $\hat{D}_s, \hat{D}_a \approx 2$, $\hat{D}_d \approx 1$, and $\hat{\tau}_s, \hat{\tau}_a \approx 1$, so the results seem reasonable.

size scaling, then we should observe a constant powerlaw coefficient $\tau_s(L) = \tau_s$, $\tau_a(L) = \tau_a$, $\tau_d(L) = \tau_d$ as well as the relationships $c_s(L) = k_s L_s^D$, $c_a(L) = k_a L_a^D$, and $c_d(L) = k_d L_d^D$ for some constants $k_s, k_a, k_d, D_s, D_a, D_d$.

To assess whether this is plausible, consider the pooled likelihood. Let $\mathbf{L} = (L_1, L_2, \cdots, L_K)$ be the set of lattice sizes simulated, and let $\mathbf{x}^j = (x_1^j, \cdots, x_n^j)$ be the set of simulated statistics from lattice size L_j.

$$\mathcal{L}(\mathbf{x}_1, \cdots \mathbf{x}_K; \mathbf{L}, \tau, k, D) = \prod_{j=1}^{K} \mathcal{L}(\mathbf{x}_j; \tau, c = kL_j^D, x_{max} = x_{max}(L_j))$$

The full dataset can be used to fit τ, k, and D using maximum likelihood, and the fit can be compared against the individual fits. If the pmf is well-specified and the system obeys finite size scaling, the fits should not significantly degrade, and we should observe that $\log \hat{c} = \log \hat{k} + \hat{D} \log L$. In other words, the pooled fit should give a trendline on the graph of $\log \hat{c}$ vs $\log L$. While the estimates in Table 4.2 appear reasonable, the results in Figure 4.4 show that the pooled parameters do not fully capture the behaviour of the pmf. In particular, $\hat{\tau}$ does not appear to be constant for either avalanche size nor area, and the fitted line on the graph of $\log \hat{c}$ vs $\log L$ is not a great fit. This suggests the estimates of $\hat{k}$ and $\hat{D}$ are smoothing over the excess error introduced by assuming a constant $\hat{\tau}$.

4.3 Stretched Exponentially Truncated Power Law

Based on the fits of the previous section, more flexibility is needed in the model to capture the steep tail decays, especially for the theoretically bounded avalanche area and length, and for small lattice sizes. We extend the exponentially truncated power law studied above with a stretching parameter β, which controls the steepness of the tail decay. We consider both the unpooled and pooled models.

The pmf is:

$$p(X = x; \tau, c, \beta, x_{max}) = \frac{1}{C_{\tau,c,\beta,x_{max}}} x^{-\tau} \exp\left(-\left(\frac{x}{c}\right)^{\beta}\right)$$

Where $C_{\tau,c,\beta,x_{max}}$ is the normalising constant:

$$C_{\tau,c,\beta,x_{max}} = \sum_{k=1}^{x_{max}} k^{-\tau} \exp\left(-(k/c)^{\beta}\right)$$

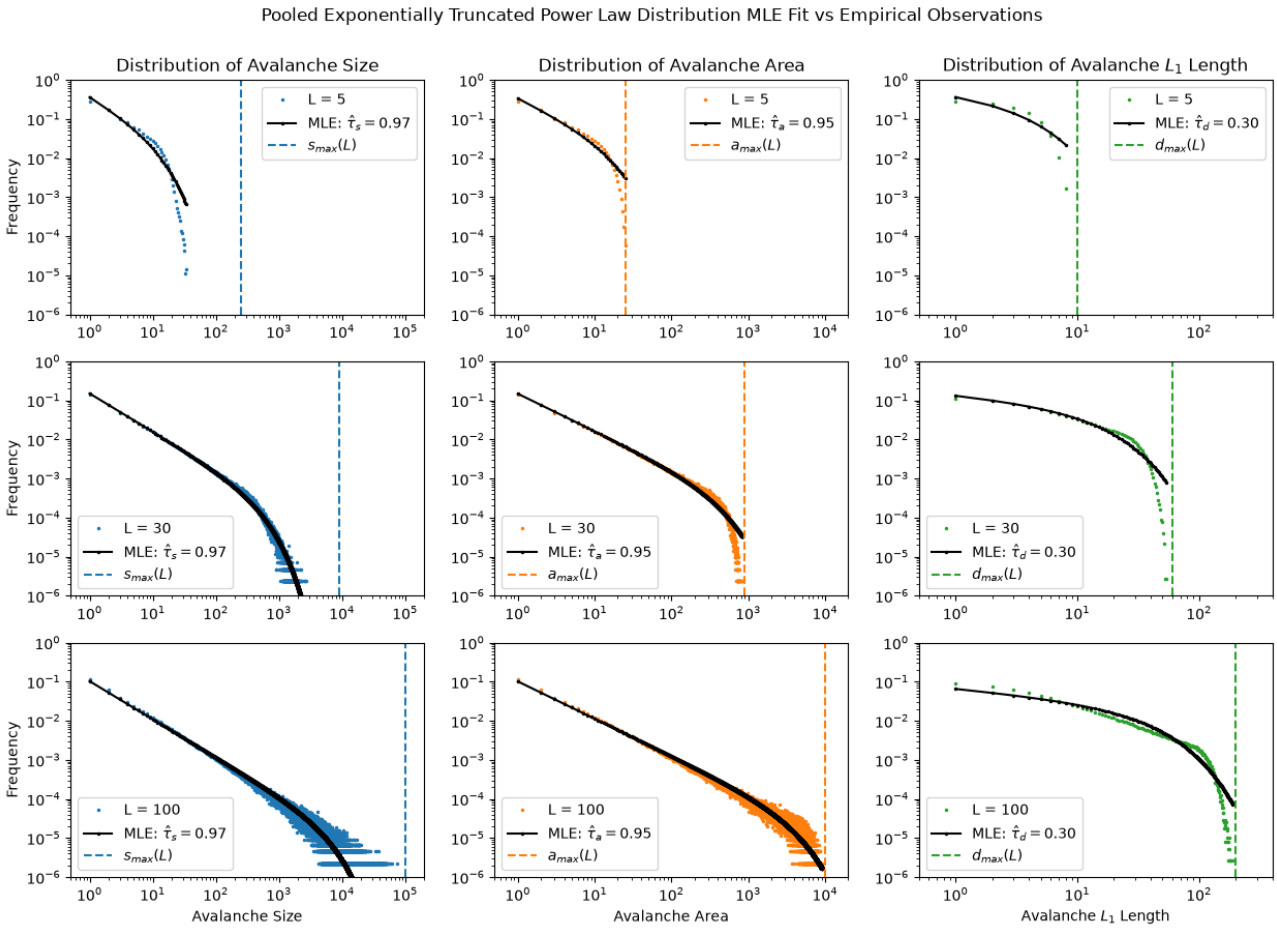


Figure 4.5: Pooled exponentially truncated power law distribution MLE fit against the simulated data, for select lattice sizes $L = 5, 30, 100$. The full simulation fits are depicted in Appendix Figure A.4. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, and the right column shows avalanche L_1 distance. The dashed vertical lines show the fixed parameter x_{max}.

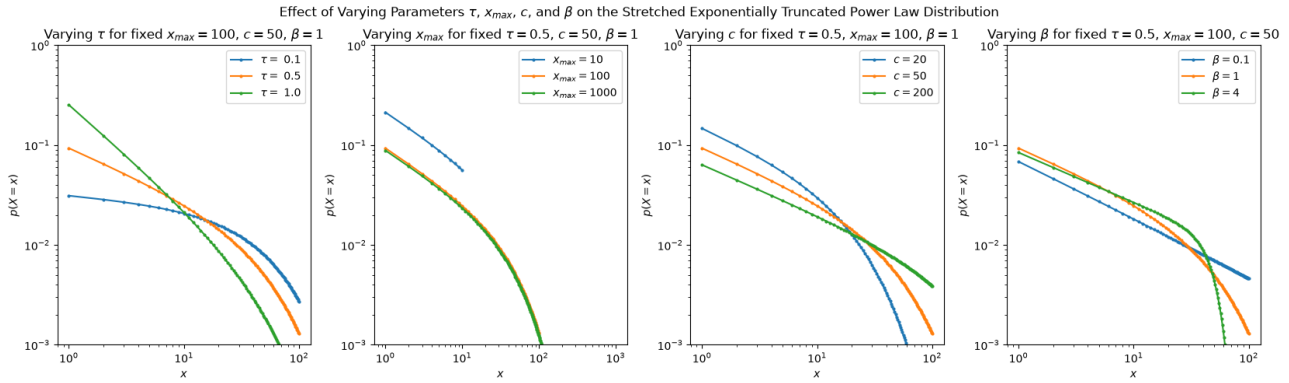


Figure 4.6: Visualisation of the stretched exponentially truncated power law probability mass function on log-log axes under different parameter combinations. Each point represents a discrete value and corresponding probability mass, with a line connecting sequential points for illustration purposes. The left three plots are similar to Figure 4.1 as they keep $\beta = 1$. The right plot shows the effect of increasing β while keeping τ, x_{max}, and c constant, which results in steeper decay.The orange line is identical in all plots, corresponding to parameters $\tau = 0.5$, $x_{max} = 100$, $c = 50$, and $\beta = 1$.

The shape of the pmf under varying parameter values can be observed in Figure 4.6. Of particular interest is that β can impose a rapid transition from linearity to decay in the log-log plot, which we especially observe in the avalanche area distribution.

The unpooled likelihood and log-likelihood are:

$$\mathcal{L}(\mathbf{x}; \tau, c, x_{max}, \beta) = \prod_{i=1}^{n} \frac{1}{C_{\tau,c,\beta,x_{max}}} x_i^{-\tau} \exp\left(-\left(\frac{x_i}{c}\right)^{\beta}\right)$$

$$\log \mathcal{L}(\mathbf{x}; \tau, c, \beta, x_{max}) = -n \log C_{\tau,c,\beta,x_{max}} - \tau \sum_{i=1}^{T} \log x_i - \frac{1}{c^{\beta}} \sum_{i=1}^{T} x_i^{\beta}$$

As before, we can extend this model under the finite size scaling hypothesis to be of the form

$$p(X = x; \tau, L, k, D, \beta) \propto x^{-\tau} \exp\left(-\left(\frac{x}{kL^D}\right)^{\beta}\right)$$

Where we assume $\tau(L) = \tau$, $\beta(L) = \beta$, as well as the relationship $c(L) = kL^D$,for some constants k, D for each statistic.

The pooled likelihood is:

$$\mathcal{L}(\mathbf{x}_1, \cdots \mathbf{x}_K; \mathbf{L}, \tau, k, D, \beta) = \prod_{j=1}^{K} \mathcal{L}(\mathbf{x}_j; \tau, c = kL_j^D, \beta, x_{max} = x_{max}(L_j))$$

The unpooled fits in Figure 4.7 appear to model the pmfs very closely, capturing the strong decay associated with small L and the area hard cutoff L^2. However, they still do not fully capture the behaviour of the avalanche L_1 length, which still has tail behaviour that is visually unexplained by the MLE fit.

The pooled fit reveals a very interesting difference between avalanche size and avalanche area. Examining Figure 4.8, while avalanche area could plausibly have $\beta_a(L) = \beta_a$, β_s has a clear L-dependence. As L increases, the soft cutoff c increases, but the steepness of the decay β decreases. This makes sense, as repeated toppling of the same locations are interrupted by loss at the boundaries, but when the lattice size increases, more grains are retained in the interior for longer, allowing repeated toppling. This is not the case for area; for any L, the avalanche area cannot exceed L^2, and therefore a sharp decay is needed near the cutoff point.

This distinction is clear in the plots of $\log \hat{c}$ vs $\log L$ in Figure 4.8 as well. While the avalanche area shows a well-fitted trendline for the pooled estimators, the avalanche size line does not appear to fit the estimates as well and is slightly offset from the trend in the unpooled estimators. This is because the estimates for k and D adapt to reduce the error of setting a fixed β_s. Because area can plausibly have a fixed β_a, this effect is minimised.

L	$\hat{\tau}_s(L)$	$\hat{c}_s(L)$	$\hat{\beta}_s(L)$	$\hat{\tau}_a(L)$	$\hat{c}_a(L)$	$\hat{\beta}_a(L)$	$\hat{\tau}_d(L)$	$\hat{c}_d(L)$	$\hat{\beta}_d(L)$
5	0.93	16.35	2.89	0.95	16.63	3.74	0.20	5.15	3.54
10	0.94	58.46	2.20	0.96	57.94	3.68	0.35	10.19	3.31
15	0.95	132.94	1.89	0.97	127.99	3.70	0.44	15.36	3.26
20	0.96	240.46	1.67	0.98	227.45	3.70	0.50	20.50	3.18
30	0.98	585.36	1.47	1.00	523.71	3.84	0.58	31.09	3.15
40	0.99	1105.57	1.35	1.01	949.78	3.89	0.64	41.66	3.07
50	1.00	1801.78	1.24	1.02	1502.50	3.91	0.67	52.02	2.99
65	1.01	3245.28	1.13	1.03	2594.20	3.95	0.71	67.22	2.83
80	1.02	5202.40	1.06	1.03	3992.73	3.95	0.75	83.47	2.83
100	1.03	8802.59	1.02	1.04	6348.53	4.11	0.77	103.50	2.66

Table 4.3: Unpooled MLE power-law slopes τ, soft cutoffs c, and stretching coefficient β obtained for varying lattice sizes L.

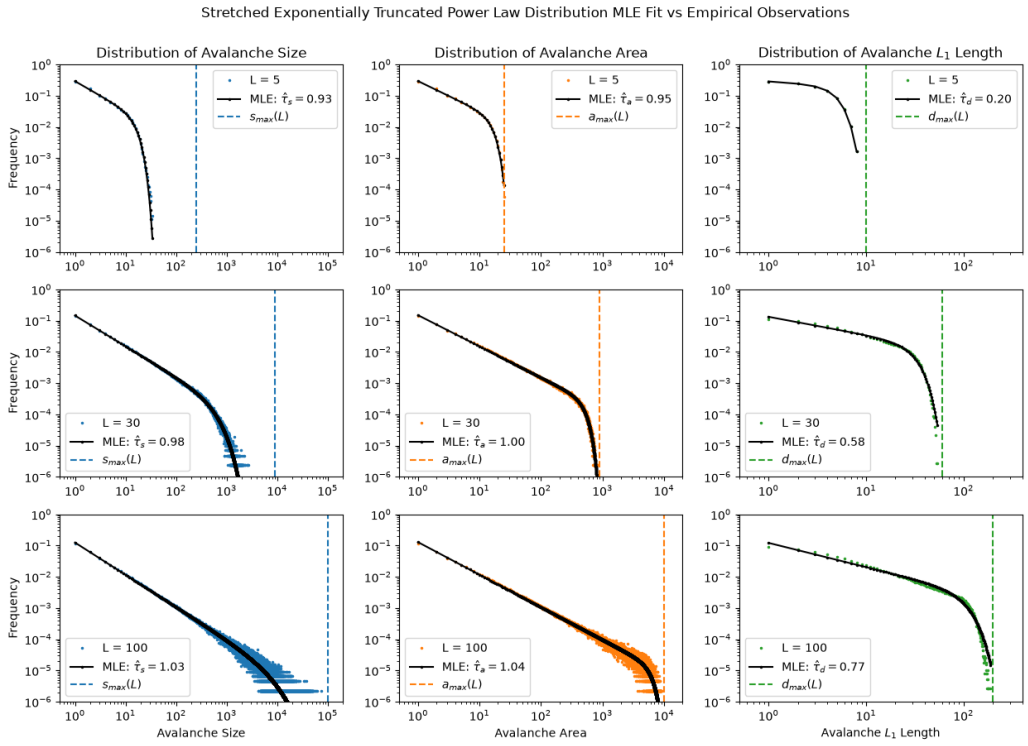


Figure 4.7: The stretching parameter β allows for a drastically better MLE fit to the data, especially for avalanche size and area at small lattice sizes. Select lattice sizes $L = 5, 30, 100$ are depicted - the full data fits are shown in Appendix Figure A.5. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, and the right column shows avalanche L_1 distance. The dashed vertical lines show the fixed parameter x_{max}.

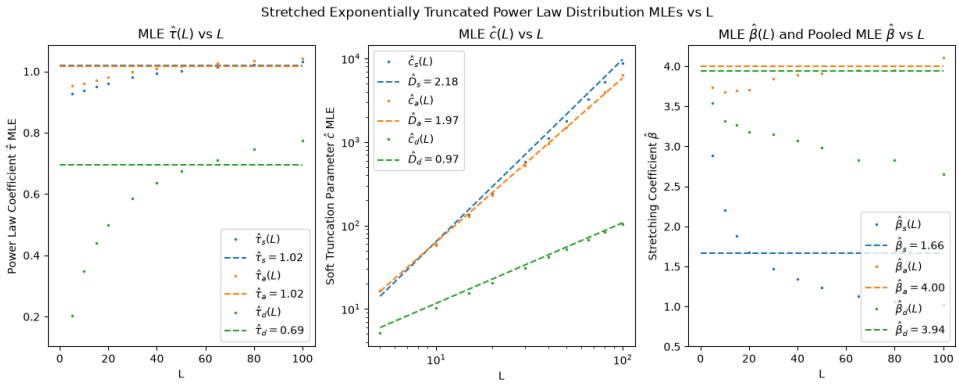


Figure 4.8: A comparison of the unpooled estimates of the parameters (points) with the pooled parameter estimates (dashed lines). If the pmf form and the finite size scaling hypothesis are appropriate, we expect the dashed line to approximate the trend of the points. Based on the above, the pooled model for avalanche area is reasonable, while evidence suggests avalanche size and length may have an $L-$dependent stretching parameter β. The power law coefficient for length also appears to be $L-$dependent.

	$\hat{\tau}$	$\hat{k}$	$\hat{D}$	$\hat{\beta}$
Size s	1.02	0.42	2.18	1.66
Area a	1.02	0.69	1.97	4.00
L_1 Length d	0.69	1.26	0.97	3.94

Table 4.4: Pooled MLE coefficients for finite size scaling assuming the stretched exponentially truncated power law distribution. We expect $\hat{D} \approx 2$ and $\hat{\tau} \approx 1$ for both avalanche area and size, so the results seem reasonable.

Comparing the unpooled MLE fits in Figure 4.7 with the pooled version in Figure 4.9, the pooled fit appears appropriate for avalanche area. Some worsening is expected, as the degrees of freedom for the fit drop, but it is not too noticeable. By contrast, the fits for avalanche size and L_1 length degrade visibly.

The above is consistent with previous work [7], which found that while avalanche area obeys finite size scaling, toppling size has multifractal scaling and is not consistent with finite size scaling. The fractal dimension $D_a \approx 2$ found here matches the estimation in [7], however the power law coefficient is lower than their estimate; we found $\tau_a \approx 1.02$ while they obtained $\tau_a = 1.2$. However, it is consistent with Bak et al.'s original paper [1].

L_1 length differs from both size and area in that its estimated power law coefficient τ_d appears to have a clear L-dependence, as well as its estimated stretching parameter β_d. Surprisingly, the pooled fit for β_d is larger than any individual fit. These signs indicate that the model may be mis-specified for length; even the manually-truncated data did not appear to scale fully as a power law.

The scaling exponent estimates $\hat{D}_d \approx 1$ and $\hat{D}_s \approx 2.2$ seem reasonable based on what they represent; length is a 1D quantity, and size can exceed L^2 through repeated topplings. However, evidence suggests these results may not be fully representative, as discussed previously.

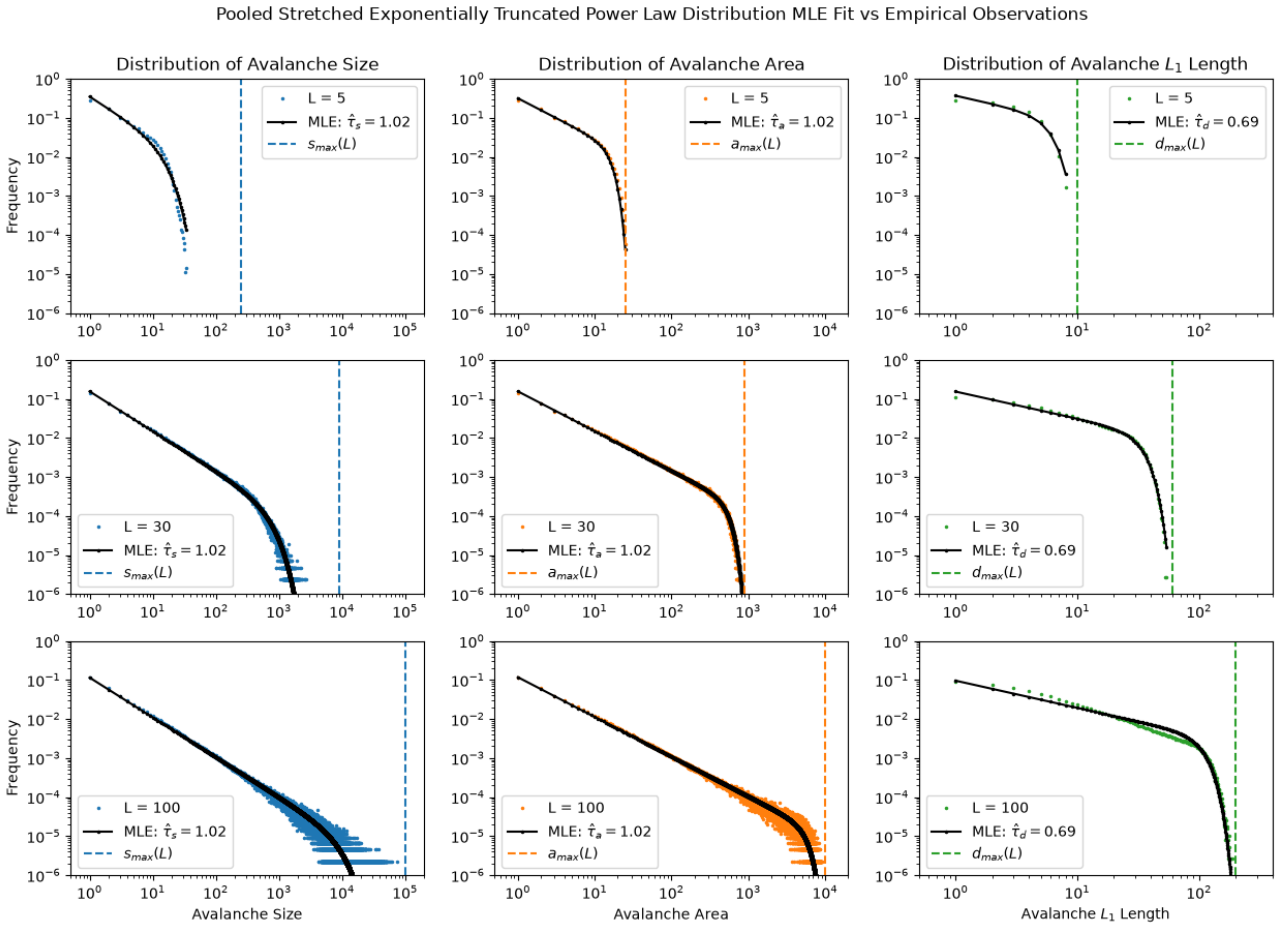


Figure 4.9: The pooled MLE fits avalanche area well, suggesting this form of finite size scaling is appropriate. Avalanche size and length appear to have more degraded fits compared with the unpooled fit in Figure 4.7. Select lattice sizes $L = 5, 30, 100$ are displayed - the full data fits are depicted in Appendix Figure A.6. MLEs are drawn as black points with a connecting line drawn for illustration purposes. The left column shows avalanche size, the centre column shows avalanche area, and the right column shows avalanche L_1 length. The dashed vertical lines illustrate the bounding parameter x_{max}.

5

Conclusion

The study found that avalanche size and area are well-modelled by the distribution family

$$p(x) \propto x^{-\tau} e^{-\left(\frac{x}{c}\right)^{\beta}}$$

Avalanche area appeared to obey finite scaling with $c = kL^D$, and we obtained estimates of the scaling coefficient $\tau_a \approx 1.02$ and fractal dimension $D_a \approx 1.97$. The latter is consistent with previous work [7], which suggests $D_a \approx 2$, but the former is somewhat lower than their calculation $\tau_a = 1.2$. Further analysis is required to obtain confidence intervals for the estimated parameters, to test whether $\tau_a = 1.2$ is plausible.

While we obtained sensible estimates $D_s \approx 2.2$ and $D_d \approx 1$, reflecting the 2D and 1D nature of the statistics, the finite size scaling fit was visually less convincing than for avalanche area. For avalanche size and length, the stretching coefficient β varied substantially depending on the lattice size L. While the power law coefficient remained relatively constant for avalanche size, $\tau_s \approx 1.02$, it was highly L-dependent for L_1 distance and exhibited an increasing trend.

5.1 Further Study

Further study is needed to quantify the fits obtained. Additional computation can re-run the simulations and optimisation procedure performed to obtain empirical distributions for the estimated parameters for each model, which can be used to construct confidence intervals.

Likewise, bootstrap estimation techniques can provide an indication of the variance of the observed distributions, which can help inform the dependency structure of the data required to perform formal goodness-of-fit hypothesis tests.

Additional proposed distributions are required for avalanche length, as the distributions examined above appear inadequate.

5.2 Reflection

This project extended my knowledge in several ways. First, on the implementation side, I was unfamiliar with the `numba` library and had to think about how to convert my Python simulation into code that could be pre-compiled. In particular, this meant re-thinking how I implemented queues, counted avalanche sizes and areas, and required me to bring in data structure knowledge from my C programming experience into Python. I also had limited experience with `numpy` vectorisation, which I used throughout my implemented pmf and likelihood functions to calculate probabilities for an array of inputs. I also learned a lot about plot functions, from how to use axes to draw multiple plot figures, to visualising asymptotes.

Second, this project made me think carefully about how we can statistically justify certain estimation procedures, especially regression of log-log plots. While this appears correct, once I carefully considered the assumptions of the linear model, it was clearly not justifiable. I learned about various power law distributions, derived the log likelihood functions, and thought about how to adjust the distributional assumption to capture variation that was not well-explained by the previous model, such as introducing the exponential truncation and stretching coefficients. I also had to justify the estimation technique from dependent samples using the Markov Chain formulation.

Finally, I increased my understanding of the abelian sandpiles model and self-organised criticality. I drew connections between my statistical models and finite size scaling, and thought about how I could assess finite size scaling by imposing restrictions on my model parameters. When the results showed that avalanche size and area differ in their pooled parameter estimation, I proposed the interpretation that avalanche size should decay more slowly as L increases, as opportunities for repeat topples increase.

Appendix A

Extended MLE Fit Plots

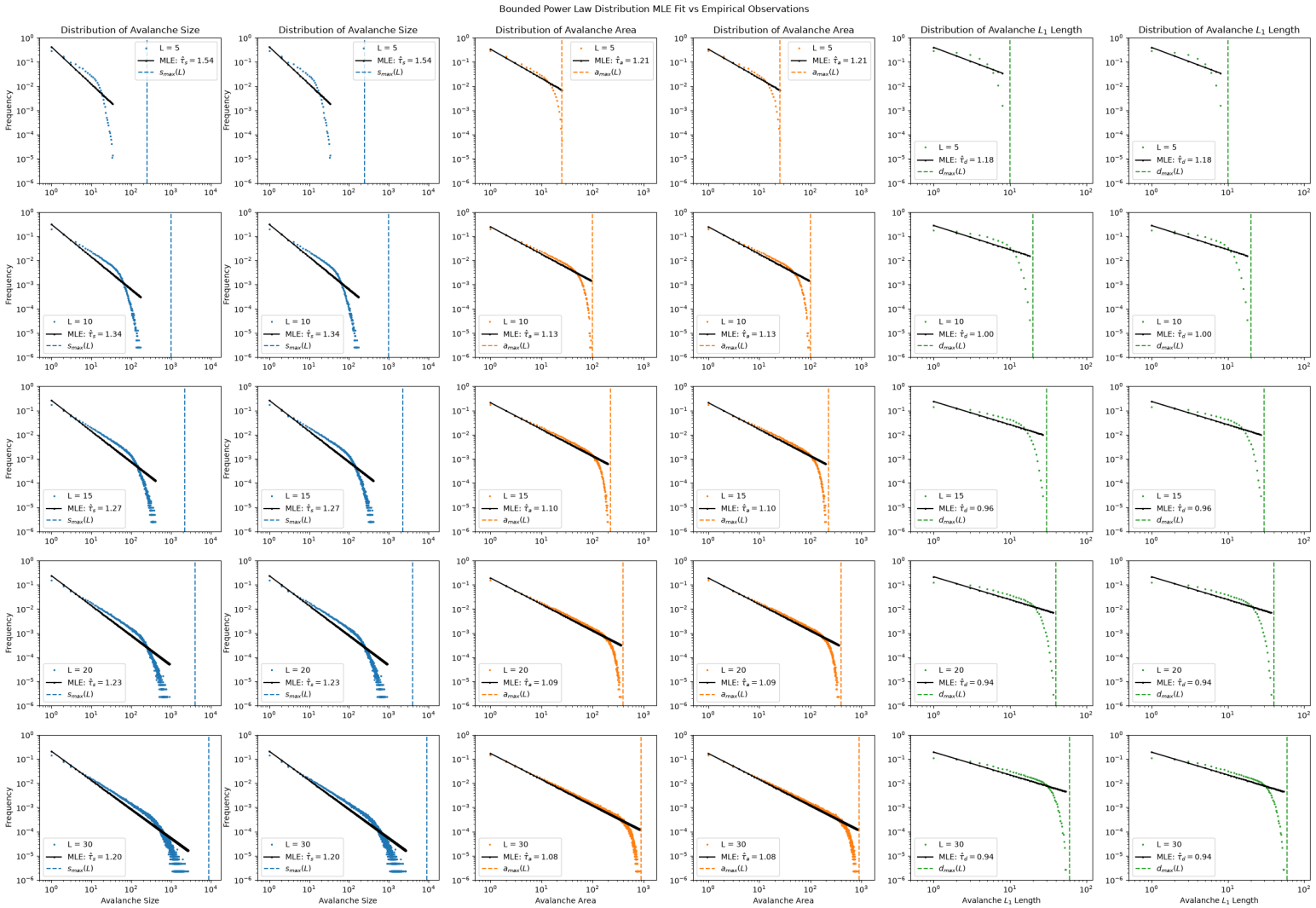


Figure A.1: Extended version of Figure 3.3 for all simulated lattice sizes.

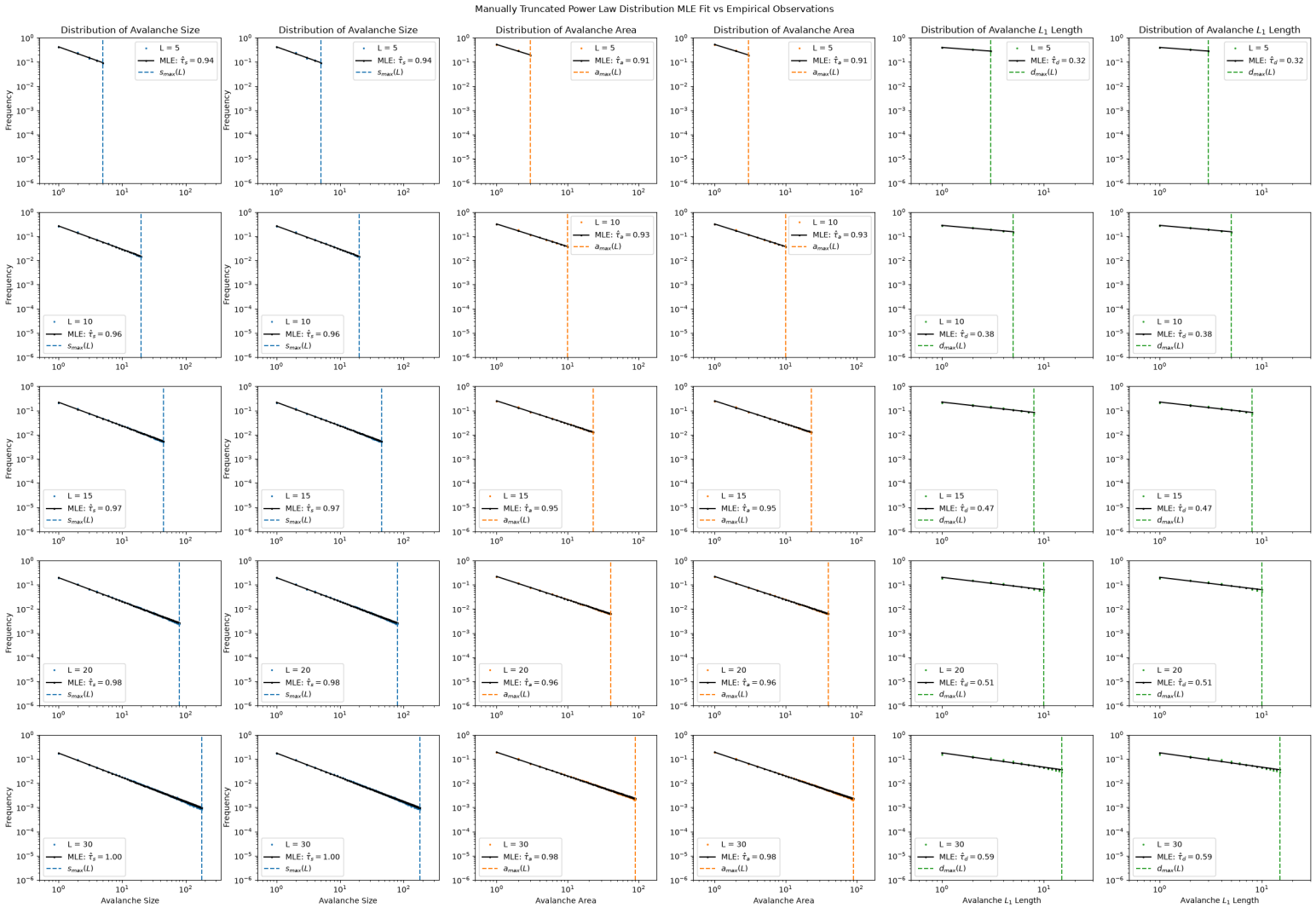


Figure A.2: Extended version of Figure 3.5 for all simulated lattice sizes.

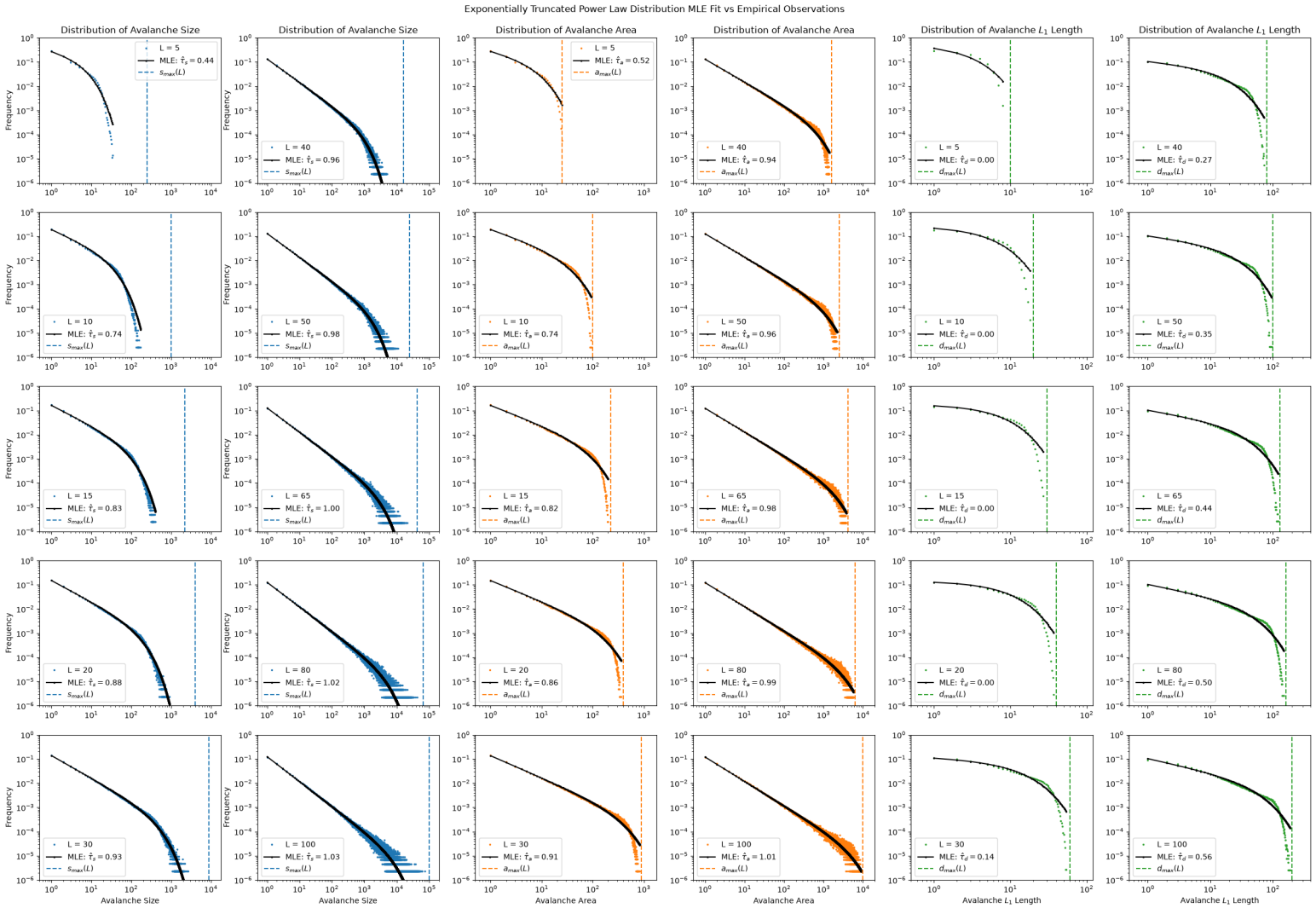


Figure A.3: Extended version of Figure 4.3 for all simulated lattice sizes.

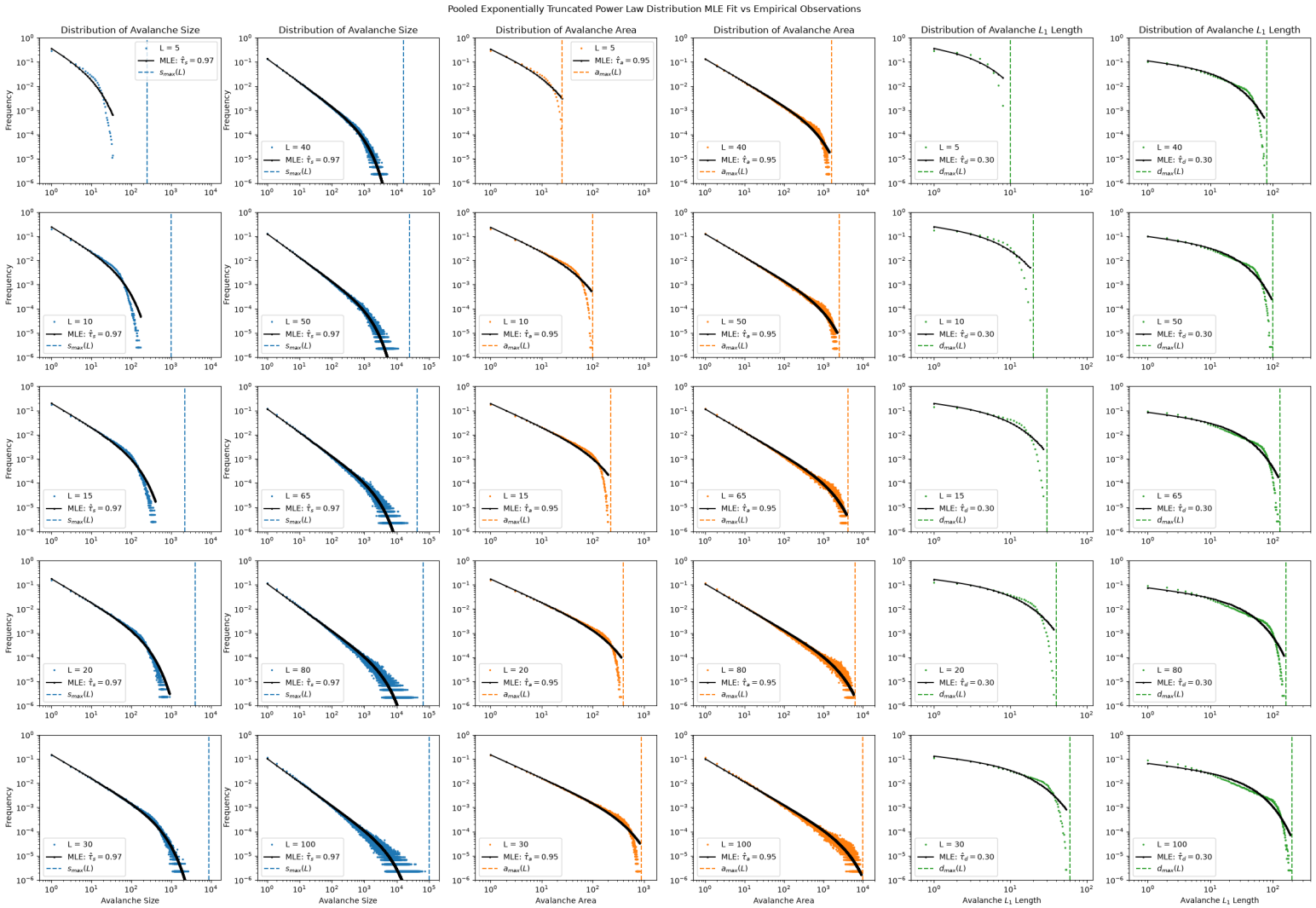


Figure A.4: Extended version of Figure 4.5 for all simulated lattice sizes.

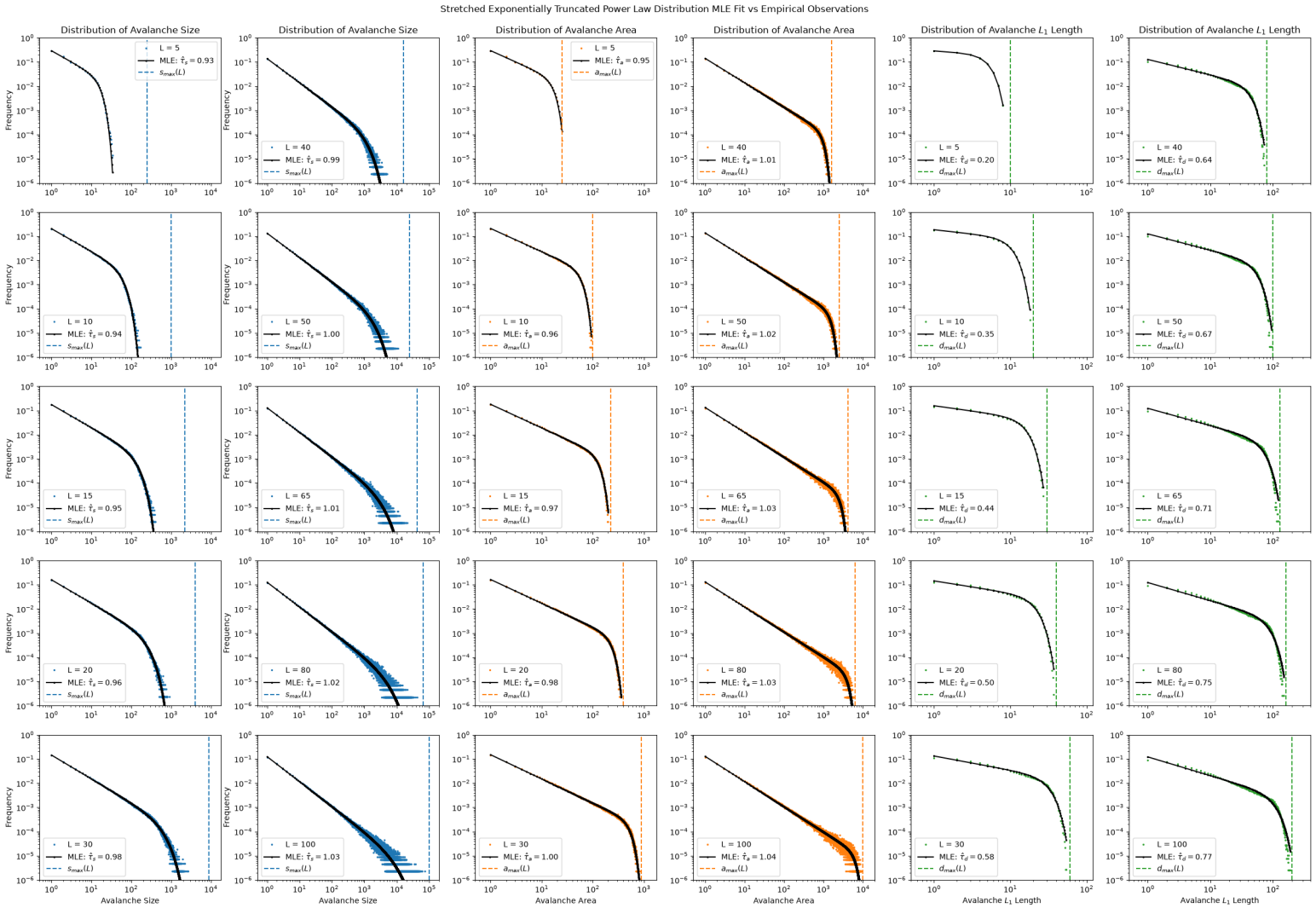


Figure A.5: Extended version of Figure 4.7 for all simulated lattice sizes.

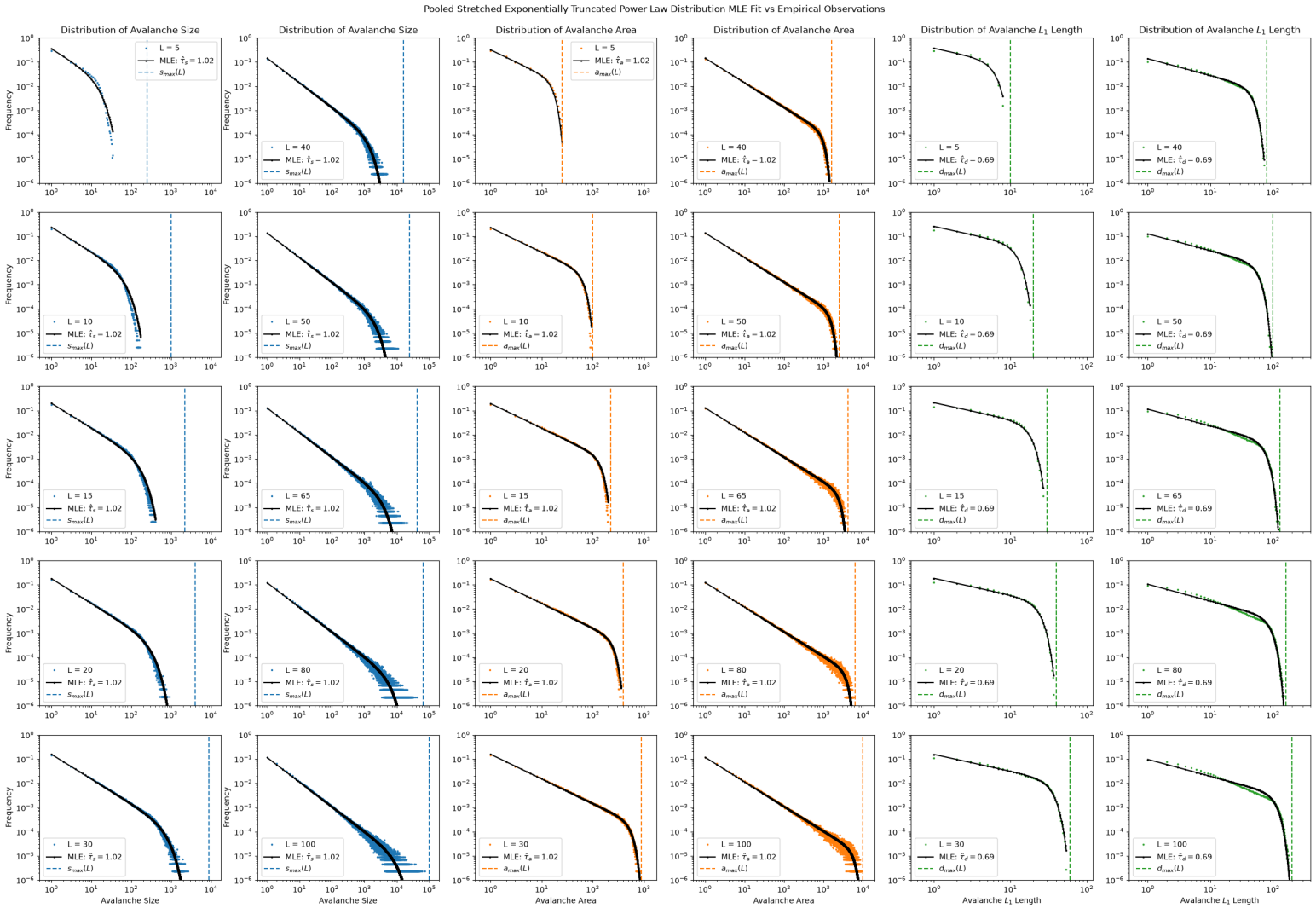


Figure A.6: Extended version of Figure 4.9 for all simulated lattice sizes.

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